IVB Model

Initial Balance breakout with volume-profile location & order-flow confirmation

Design, trade logic, module contracts, inputs & outputs · package `strategies/ivb_model/` *· updated 2026-07*

0. What the strategy is

The **IVB (Initial-balance Volume-profile Breakout)** model is a directional intraday futures strategy. Its thesis is simple: the market spends the first part of the session discovering a fair-value range (the *initial balance*); when price finally commits beyond that range it reveals who won the day's most important auction; the highest-probability entry is not the breakout itself but the **first pullback that fails** — where price returns toward value, the counter-move runs out of aggressive participants, and the order flow shows that failure directly. The model was originally inspired by Fabio Valentini's IVB concept and has since been rebuilt around a modular, plug-in package.

The design separates into two layers. A **directional framework** (define the initial balance, map the value area, detect a breakout, wait for a retest) is fixed and shared. An **entry-confirmation layer** is pluggable — seven interchangeable order-flow patterns, each of which can be switched on or off independently — followed by a pluggable **risk layer** of three interchangeable stop/target scripts. Everything that differs between research variants lives in those two plug-in registries; the spine never changes.

The three questions every trade must answer

Layer	The question it answers	How the model answers it
1 · Direction	Who won the most important battle of the day?	The first post-IB bar to **close** beyond the initial-balance high (long) or low (short).
2 · Location	Where is the reload / value zone the pullback must fail at?	VAH / POC / VAL, built from the IB's traded-volume-at-price profile.
3 · Confirmation	Is the counter-move actually failing at that zone right now?	One of seven order-flow entry patterns (Section 4) fires inside the value area.

> **How to read this document**
>
> Sections 2–7 explain the **logic** — what each stage is trying to detect in the market and why — with the exact numeric rules the code enforces, but without requiring you to read the code. Sections 8–12 are the **contract**: parameters, inputs, outputs and known limits. The companion `strategies/ivb_model/CLAUDE.md` is the terse code-map for engineers; this PDF is the long-form rationale.

1. Package structure & cross-connection

IVB is a multi-file Python **package** (a folder with an `__init__.py`), not a single-file strategy. The backtester discovers it by folder name and calls its `run(folder_path, start_date, end_date, params)` entry point, exactly as it would call a one-file strategy. Internally the work is split so that each concern lives in one place and the two plug-in registries (entries and risk) can grow without touching the spine.

File	Provides	Role / contract
`__init__.py`	`run(...)`, re-exports `PARAMS`, `PARAM_SECTIONS`	Globs the day files in range, resolves the optional indicators folder, calls `process_day` per day, returns the `OUTPUT_COLUMNS` frame.
`params.py`	`PARAMS`, `PARAM_SECTIONS`, `OUTPUT_COLUMNS`	The full default parameter dict, its UI grouping, and the output schema.
`core.py`	`detect_breakout`, `detect_retest`, `find_entry`, `process_day`	Orchestrates one trading day: breakout/retest detection, the entry dispatcher, the flip loop, and the risk-script dispatch.
`profile.py`	`compute_ivb_profile(ib_bars)`	Peak-based volume profile → `(poc, vah, val)` from the IB's `tick_volume` (Section 5).
`baselines.py`	`build_rolling_baseline`, `build_passive_baseline`, `build_cvd_change_baseline`	The three day-level rolling baselines that absorption / passive / CVD grading normalise against (Section 3).
`absorption.py`	`is_absorption_candle`, `find_absorption_trigger`	Shared wick-absorption grading read from `tick_volume`; reused by several finders.
`entries/`	7 finders + `FINDER_REGISTRY`	One module per entry pattern; each exposes `find_entry(**shared)` returning a 5-tuple (Section 4).
`risk/`	3 scripts + `RISK_REGISTRY`	Self-contained stop/target placement + fill simulation; selected 1-based by `risk_script` (Section 6).

Per-day call chain

`run()` loops the day files; all real work for a single session happens in `process_day(session, params, ind_df)`:

```
run()  ->  for each YYYY-MM-DD.parquet in [start, end]:
  process_day(session, params, ind_df):
    1. slice RTH 09:30-16:00 NY          (need >= ib_minutes bars)
    2. IB high/low from first ib_minutes bars   (abort if range <= 0)
    3. compute_ivb_profile(ib_bars)      -> poc, vah, val      [profile.py]
    4. build rolling + passive baselines (long & short)       [baselines.py]
    5. from indicators (if present): CVD series + CVD-change std,
       and the tick-VWAP deviation bands for the risk layer
    6. detect_breakout(post_ib)          -> direction, breakout_pos
    7. loop (<= max_flips):
         detect_retest()                 -> retest_pos  (else: no trade)
         post_retest = [breakout bar] + [retest .. retest+entry_window]
         find_entry() -> earliest entry across enabled finders wins
            entry      -> break out of loop
            invalidate -> flip direction, resume after invalidation, re-detect
    8. RISK_REGISTRY[risk_script-1](...) -> trade dict | None
    9. attach trade_type + notes (JSON)
```

2. The directional framework (the spine)

This layer is deterministic and identical for every entry and risk plug-in. It turns a day of candles into a direction, a value area to fade toward, and a bounded window in which the confirmation layer is allowed to look for a trade.

Step 1 — Initial Balance

Take the first `ib_minutes` bars of the regular session (default 30 → 09:30–09:59 NY). The IB high is the max of those highs, the IB low the min of those lows. These two prices are the walls of the day's opening auction. If the range is zero the day is skipped. The bar immediately after the IB (10:00 with the default) is the first candidate breakout bar.

Step 2 — Value area (location)

From the same IB bars the model builds a volume-at-price profile from `tick_volume` and extracts three levels: the **POC** (point of control — the most-traded price), the **VAH** and **VAL** (the top and bottom of the 70% value area). These define "where value is" and become the reload zone a healthy breakout should pull back into, plus the line whose violation kills the idea. They are computed once and reused across every flip. The peak-based algorithm that produces them is detailed in Section 5.

Step 3 — Breakout (direction)

Scan post-IB bars for the first one whose **close** is beyond a wall: close > IB-high → **long**, close < IB-low → **short**. Whichever happens first sets the direction (on the same bar, long wins the tie). A wick that pokes through but closes back inside is *not* a breakout — the model insists on a committed close, which filters the stop-runs that plague naive opening-range systems.

Step 4 — Retest

After the breakout, wait up to `retest_window` bars for price to pull back into value: for a long, a bar's **low** ≤ **VAH**; for a short, a bar's **high** ≥ **VAL**. That pullback is the setup — the market offering a second, lower-risk chance to join the winning side at a better price. The window handed to the confirmation layer, `post_retest`, is the **breakout bar** (kept as index 0 for reference) followed by the retest bar and the next `entry_window` bars. If no retest arrives in time, there is no trade that day.

Step 5 — Invalidation & direction flipping

The whole idea is invalidated when price closes back *through the far side of value*: a long dies on a **close < VAL**, a short on a **close > VAH**. Rather than simply abandon the day, the model treats a decisive failure as information — it **flips**: reverses the intended direction, resumes scanning from just after the invalidation bar, and re-detects a breakout in the new direction. This repeats up to `max_flips` times (default 4), always reusing the same IB profile. It is deliberately conservative: if the re-detected breakout does not actually point the flipped way, the day is dropped rather than forced.

> **Why fade toward value instead of chasing the break?**
>
> A breakout that is real gets *defended* on the retest: the players who pushed price out of balance step back in to stop it re-entering. That defense is visible in order flow as absorption, resting size, or a delta divergence. Fading the failing pullback lets the model enter with a tight, logical stop (the value-area edge) and a clear invalidation, instead of buying strength at the worst price.

3. Absorption grading & the rolling baselines

"Absorption" is the model's core order-flow primitive: aggressive traders keep hitting a level, yet price refuses to move through it — someone is quietly filling all of that aggression with passive orders. Four of the seven finders lean on the same grading routine so the definition is consistent.

What makes a candle an absorption candle

`is_absorption_candle` calls a bar absorption when **both** hold on the defended side (the lower wick for a long, the upper wick for a short):

- **Shape:** the defended wick is at least `wick_threshold` of the bar's total range (default 40%) — price was pushed into that zone but rejected, leaving a long tail.
- **Flow:** somewhere inside that wick, a single price level shows counter-trend aggressive volume ≥ `baseline × absorption_mult` (default 2×). For a long that is aggressive *selling* being soaked up at the lows; for a short, aggressive *buying* soaked up at the highs.

When a candle qualifies, `find_absorption_trigger` reports the first `(price, volume)` level that crossed the threshold, which is written into the trade notes for later inspection. The multiplier and wick threshold are per-finder: the consecutive and passive finders pass their own values in, so "absorption" can be graded more or less strictly per pattern without changing this shared routine.

The rolling baselines (what "big" is measured against)

Raw volume is meaningless without a local reference. Each baseline is a rolling, day-level average computed over `absorption_baseline_window` bars (default 20), and every one of them **skips the first five minutes** of RTH (baselining starts 09:35) because the opening bars are abnormally heavy and would inflate the reference. During warm-up the baseline is undefined, so the earliest candles simply cannot be graded.

Baseline	What it averages	Used by
`build_rolling_baseline`	Per-tick aggressive volume = buy/sell volume ÷ bar range in ticks. Returns a **sell** baseline (for longs) and a **buy** baseline (for shorts).	absorption_delta, consecutive, passive size-only
`build_passive_baseline`	The largest **resting** order size on the defended side per bar (bids below the open for longs, asks above it for shorts). Sparse — only bars with valid passive data count.	passive size-only, passive wall
`build_cvd_change_baseline`	Rolling **standard deviation** of bar-to-bar CVD change — the session's own order-flow volatility, used to z-score divergences.	both CVD-divergence finders

A fourth, finder-local baseline lives inside the two-bar module: `build_two_bar_baseline` averages the defended-side per-tick volume of *merged two-bar pairs* preceding the pattern, because that finder grades a synthetic two-candle unit rather than a single bar.

4. The seven entry finders

Every finder shares one signature and returns the same 5-tuple `(entry_ts, entry_price, invalidation_ts, entry_notes, trade_type)`. The dispatcher (`find_entry`) runs every finder switched on by the `valid_entries` flag string over the identical `post_retest` window and shared baselines, then picks the **earliest entry**. If no finder entered but one signalled invalidation, the earliest invalidation is returned and drives a flip. Two universal rules hold across all seven:

- **Entry price is always the OPEN of the bar after the confirming candle** — the signal must fully close before the model commits, so there is no intrabar look-ahead.
- **Whole-trade invalidation short-circuits every finder**: a close through VAL (long) / VAH (short) at or before a candidate bar returns an invalidation instead of an entry.

The `valid_entries` string is one bit per finder in registry order (1 = on). The default `"1111100"` enables the five order-flow patterns and leaves the two CVD-divergence finders off (they require the optional indicators dataset). Bit order is fixed:

#	trade_type	The pattern, in words	Key params
1	`absorption_delta`	An absorption candle at the zone, then a **confirming** candle in the trade direction with a body ≥ threshold and delta pushed past ±threshold. The classic "absorb then reverse".	wick_threshold, absorption_mult, delta_threshold, body_threshold
2	`consecutive_absorption`	N absorption candles (default 2) clustering within a few ticks of the same level and body midpoint — repeated defense of one price. **No** delta confirmation needed; the repetition is the signal.	consec_abs_n, consec_abs_mult, consec_abs_ticks, consec_wick_threshold
3	`two_bar_absorption`	A tight reversal **pair** (both with small defended wicks) merged into one synthetic candle, graded against a two-bar baseline, then a confirming candle. Catches absorption that splits across two bars.	two_bar_wick_ticks, two_bar_abs_mult
4	`passive_absorption_size_only`	A genuinely **large resting order** on the defended side (raw size ≥ baseline×mult) *and* absorption on the same candle, then a confirming candle. Passive intent + realised absorption together.	passive_size_order_mult, passive_size_absorption_mult, passive_size_wick_threshold
5	`passive_wall`	A **cluster** of big resting orders (default 3) stacked within a few ticks of one level. No absorption candle and no delta — the wall of stacked liquidity is itself the signal.	passive_wall_n, passive_wall_mult, passive_wall_ticks
6	`cvd_divergence_absorption`	A CVD divergence read as absorption: price could *not* make a new extreme even though cumulative delta pushed further — the extreme was absorbed. **Needs CVD.**	cvd_pivot_k, cvd_min/max_separation, cvd_wick_tolerance_ticks, cvd_min_score
7	`cvd_divergence_exhaustion`	The mirror, read as exhaustion: price *did* make a new extreme but cumulative delta failed to follow — the move ran out of fuel. **Needs CVD.**	(same CVD-divergence params)

The confirming candle (patterns 1, 3, 4, 6, 7)

Most finders separate *detection* from *confirmation*. After spotting absorption or a divergence they scan up to `entry_after_absorption` bars (default 5) for a confirming candle: correct colour, body ≥ `body_threshold` of range, and signed order-flow delta past `±delta_threshold`. Only then — on the *next* bar's open — is the position taken. Patterns 2 and 5 skip this step: repeated absorption and a stacked wall are treated as self-confirming, so they enter on the bar right after the pattern completes.

How the two CVD-divergence finders work

Both build swing pivots with a left-side k-bar fractal that is confirmed quickly by a single reversal candle (no waiting k bars to the right), then grade the two most-recent pivots. The distinction is purely the price-vs-delta relationship:

- **Absorption** (finder 6): the newer extreme is *lower-or-equal* (short) / *higher-or-equal* (long) than the older, yet CVD moved further into it — more aggression, no new extreme → absorbed.
- **Exhaustion** (finder 7): the newer extreme *does* exceed the older, yet CVD moved *less* — a new extreme on weaker flow → exhausted.

The CVD gap between the pivots is turned into a **z-score** by dividing by the session's own CVD-change volatility (the third baseline), and must exceed `cvd_min_score` (default 0.3) to qualify — so "divergence" is measured relative to how noisy that particular day's order flow is, not an absolute number. A freshly confirmed third pivot supersedes any pending setup, rolling the sliding pair forward. If no indicators were loaded, both finders quietly disable themselves and return all-None, never blocking the others.

> **Adding an eighth entry type**
>
> Drop `entries/my_entry.py` exposing `find_entry(**shared) -> 5-tuple`; append it to `FINDER_REGISTRY`; extend the default `valid_entries` string by one bit and add any new parameters to `params.py`. No other file changes.

5. Volume-profile algorithm (peak-based)

A naive "most-traded price" POC is fragile — one fat tick can move it. IVB uses a peak-based method so the value area sits on a genuine cluster of activity:

- Aggregate `tick_volume` across all IB bars into a `{price: total_volume}` map.
- Smooth with a 3-tick rolling average to remove single-tick spikes, then find local maxima.
- Cluster peaks within 4 ticks of each other, keep the strongest per cluster, and take the top 5.
- Refine each surviving peak to the true highest-volume *raw* tick within ±3 ticks → a POC candidate.
- Expand a 70% value area outward from each candidate; the candidate whose value area is the **tightest** (smallest VAH–VAL) wins — concentration beats sprawl.
- Finally re-pin the POC to the highest raw-volume tick *inside* the winning value area.

If fewer than three distinct price levels exist (a very thin IB), it falls back to a simple max-volume POC expanded to 70%. The result is a POC/VAH/VAL triple that is stable enough to anchor both the retest test and the stop logic.

6. Risk management — three interchangeable scripts

`risk_script` is a **1-based selector** into `RISK_REGISTRY`. Each script is fully self-contained: it owns both its stop/target placement and its own copy of the fill simulator — there is no shared module and no cross-script imports, so any script can be edited in isolation. All three receive the same inputs (the retest window, the post-entry bars, the entry, the direction, the value-area levels, and the params) and return the standard trade dict or None.

risk_script	Script	Stop placement	Target
1	`basic_risk`	VAL/VAH (`sl_type=0`) or the swing low/high up to entry (`sl_type=1`)	Fixed reward multiple: entry ± risk × `rr`
2	`zone_sl_risk`	Where the *pullback* actually bottomed/topped, snapped to the VAL/POC/VAH zone it landed in (no look-ahead — entry bar excluded)	Fixed reward multiple: entry ± risk × `zone_rr`
3	`vwap_tp_risk`	VAL/VAH or the zone logic, chosen by `sl_placement`	A tick-VWAP ±2σ / ±3σ deviation band, frozen at entry or trailed live

Stop logic in words

- **basic_risk** — the simplest: stop at the far value-area edge (VAL for longs, VAH for shorts), or optionally at the swing extreme formed up to the entry. Target is a flat multiple of that risk.
- **zone_sl_risk** — recognises that a pullback which only dipped to POC deserves a tighter stop than one that sank to VAL. It reads where the pullback's *closes* bottomed (or topped), decides which half of the value area

that was, and places the stop at that zone edge — or at the actual wick extreme if price poked beyond it. The entry bar is excluded so the stop never uses future information.

- **vwap_tp_risk** — keeps a zone/edge stop but aims the target at a statistical mean-reversion band (tick-VWAP ±2σ or ±3σ, globex or RTH session). If price is already past the chosen band at entry it **escalates**: past 2σ bumps the target to 3σ; past 3σ falls back to a plain 1:1. In `"trailing"` mode the band target moves bar-by-bar (with a pessimistic same-bar tie going to the stop); in `"now"` mode the band value is frozen at entry.

Trade fill simulation (shared three-phase shape)

Once the stop and target are set, every script simulates the fill the same way:

- **Phase 1 — up to `trade_timeout` bars:** walk forward looking for the stop or the target. If both are touched on one bar, the **target wins** the tie (an optimistic assumption). Exit reason `tp` or `sl`.
- **Phase 2 — the timeout bar:** if the position is in profit at the timeout close, bank it (`tp_timeout`). Otherwise pull the target to breakeven and the stop to the swing extreme, then keep running.
- **Phase 3 — after the timeout:** exit on the adjusted breakeven target (`tp_timeout`), the adjusted swing stop (`sl_timeout`), or at the session close (`eod`) if neither is reached.

> **Default behaviour of the timeout**
>
> The default `trade_timeout = 999` is larger than a normal RTH session (≈390 one-minute bars), so with defaults the trade almost always resolves in Phase 1 (a clean tp / sl / eod) and the breakeven-and-run machinery stays dormant. Lower the value to activate it.

exit_reason	Meaning
`tp` / `sl`	Target / stop touched within the timeout window (target wins same-bar ties).
`eod`	Session ended (or data ran out) before any level was hit — exit at the last close.
`tp_timeout`	In profit at the timeout, or the post-timeout breakeven target was hit afterwards.
`sl_timeout`	The post-timeout swing stop was hit after the breakeven adjustment.

A risk script may attach a `risk_notes` dict (e.g. `vwap_tp_risk` records `tp_type` = `tp_vwap_2` / `tp_vwap_3` / `1:1` and an `escalated` flag). It is popped in `process_day` and merged into the notes JSON so it never becomes a stray output column.

7. The optional indicators layer (CVD & VWAP)

Two of the seven finders and the third risk script use an **optional** second dataset. `__init__.py` resolves it as a sibling folder under the same `parquet/{type}/{asset}/` directory, named by the `indicators_folder` parameter, and loads the matching `YYYY-MM-DD.parquet` for each day. If the parameter is empty, the folder is missing, the daily file is absent, or the read fails, the day simply runs **without** indicators: the two CVD finders disable themselves and `vwap_tp_risk` returns no trade. Nothing else is affected.

Indicator column(s)	Consumed by	For
`cumulative_delta`	both CVD-divergence finders + `build_cvd_change_baseline`	Pivot CVD readings and the z-score volatility denominator.
`vwap_tick_{globex,rth}_std{2,3}_{up,dn}`	`vwap_tp_risk`	The ±2σ / ±3σ deviation-band take-profit targets (8 columns).

These columns are produced by `transforms/1m_advanced_indicators.py`. A third parameter, `big_trades_folder`, is declared for an upcoming feature but is **reserved** — not yet consumed by any finder.

8. Parameters

Defaults come from `params.py`; `PARAM_SECTIONS` groups them in the backtester UI. Current values:

Parameter	Default	Meaning
`ib_minutes`	30	IB duration in bars (typically 15 / 30 / 60).
`trade_timeout`	999	Bars before the timeout/breakeven logic engages (999 ≈ off).
`max_flips`	4	Max direction flips per day after invalidation.
`valid_entries`	"1111100"	One on/off bit per finder, in registry order.
`risk_script`	1	1 = basic_risk, 2 = zone_sl_risk, 3 = vwap_tp_risk.
`indicators_folder`	"ES_1m_indicators"	Sibling dataset for CVD + VWAP bands; empty = none.
`big_trades_folder`	"ES_big_trades"	Declared, reserved for upcoming use.
`retest_window`	45	Max bars to wait for a retest after breakout.
`entry_window`	25	Bars scanned for an entry after the retest.
`entry_after_absorption`	5	Bars scanned for a confirming candle after a signal.
`absorption_baseline_window`	20	Rolling window (bars) for every baseline.
`delta_threshold`	10.0	Min |volume_delta_pct| for a confirming candle.
`body_threshold`	0.5	Min body/range ratio for a confirming candle.
`wick_threshold`	0.4	Min defended-wick/range ratio (absorption_delta).
`absorption_mult`	2.0	Wick-level volume must be ≥ this × baseline.
`consec_abs_n`	2	Absorption candles required at the same level.
`consec_abs_mult`	2.0	Absorption multiplier (consecutive).
`consec_abs_ticks`	4	±tick tolerance for grouping levels (consecutive).
`consec_wick_threshold`	0.4	Wick threshold (consecutive).
`two_bar_wick_ticks`	8	Max defended wick, in ticks, on both bars of the pair.
`two_bar_abs_mult`	2.0	Absorption multiplier for the merged pair.
`passive_size_order_mult`	2.0	Raw resting size ≥ this × passive baseline.
`passive_size_absorption_mult`	1.5	Absorption multiplier (passive size-only).
`passive_size_wick_threshold`	0.4	Wick threshold (passive size-only).
`passive_wall_n`	3	Big resting orders needed to form a wall.
`passive_wall_mult`	2.0	Raw size ≥ this × baseline to count as “big”.
`passive_wall_ticks`	8	±tick tolerance for clustering wall levels.
`cvd_pivot_k`	2	Left-side bars required to qualify a pivot (fractal).

Parameter	Default	Meaning
cvd_min_separation	3	Min bars between the two pivots.
cvd_max_separation	20	Max bars between pivots (older pivot stale beyond).
cvd_wick_tolerance_ticks	2	Tick tolerance on the pivot price comparison.
cvd_min_score	0.3	Min |z-score| of the CVD divergence.
rr	1.0	Fixed risk:reward (basic_risk).
sl_type	0	0 = VAL/VAH stop, 1 = swing low/high (basic_risk).
zone_rr	1.0	Fixed risk:reward (zone_sl_risk).
sl_placement	2	1 = VAL/VAH stop, 2 = zone logic (vwap_tp_risk).
vwap_std	2	Which sigma band to target: 2 or 3 (vwap_tp_risk).
vwap_session	"globex"	VWAP band session: globex or rth.
vwap_tp_mode	"now"	Band frozen at entry (now) or trailed live (trailing).
tick_size	(auto)	Injected from ASSET_INFO; in HIDDEN_PARAMS — no UI widget.

9. Output schema

The strategy returns one row per trade with these `OUTPUT_COLUMNS`. It never stores a ticks column — the backtester derives ticks from `pnl_points` using the asset's tick metadata.

Column	Type	Description
date	date	Trade date (the day file).
direction	str	Lowercase 'long' / 'short'.
trade_type	str	Which finder fired (one of the seven names in Section 4).
entry_time / exit_time	Timestamp	Entry-bar open time / exit-bar time.
entry_price / exit_price	float	Entry-bar open / the SL, TP, or close at exit.
sl / tp	float	Stop-loss / take-profit levels used.
exit_reason	str	tp / sl / eod / tp_timeout / sl_timeout.
pnl_points	float	exit–entry (long), entry–exit (short), in points.
notes	JSON str	Context + finder + risk notes (Section 10).

10. The notes column

Per-day context is merged with finder-specific and risk-specific keys and serialised as JSON. The backtester renders it as a second metrics row. A representative long via `absorption_delta`:

```
{
  "breakout_time": "10:03",  "retest_time": "10:11",
  "flip_count": 0,
  "ivb_high": 5102.25, "ivb_low": 5094.50,
  "poc": 5097.75, "vah": 5100.25, "val": 5095.50,
  // + finder-specific keys, e.g. absorption_delta:
  "absorption_time": "10:14", "abs_baseline": 118.4,
  "trigger_price": 5096.0, "trigger_volume": 412
  // consecutive/two_bar/passive/wall/cvd add their own keys;
  // vwap_tp_risk adds tp_type + escalated
}
```

11. Data requirements (inputs)

IVB requires enriched one-minute candles from `transforms/1m_advanced.py`, currently available for **ES / NQ only** (the enriched Databento window). Plain OHLCV is not sufficient. Each day is one `YYYY-MM-DD.parquet` with a tz-aware `America/New_York` index.

Column	Required	Used for
`open` `high` `low` `close`	yes (float64)	IB range, breakout/retest, bar structure, SL/TP.
`buy_volume` `sell_volume`	yes	The per-tick rolling absorption baseline.
`volume_delta_pct`	yes	Confirming-candle delta threshold.
`tick_volume` (JSON {price:[buy,sell]})	yes	Volume profile + absorption grading.
`passive_orders` (JSON {price:[size,count]})	yes	The two passive finders + passive baseline.
`volume`	no	Present but not used directly by the core logic.

The optional indicators dataset (Section 7) adds `cumulative_delta` and the `vwap_tick_*` band columns from `transforms/1m_advanced_indicators.py`; without it the model still trades on the five order-flow finders and the two point-based risk scripts.

12. Known limits & roadmap

- **Session times are hardcoded 09:30–16:00 NY** — correct for equity-index futures, wrong for other asset classes with different RTH.
- **Enriched data coverage is thin** — tick_volume / passive_orders exist for ES/NQ over roughly one year, so walk-forward validation needs multi-year enriched data.
- **Baseline warm-up** — baselines need a full 20-bar window from 09:35, so the earliest post-open candles cannot be graded.
- **Flip conservatism** — a re-detected breakout in the wrong direction yields no trade rather than a forced entry.
- **No higher-timeframe context** — the model operates purely on one-minute bars.

Capability	Status
Peak-based volume profile + POC refinement inside the value area	done
Absorption+delta, consecutive, two-bar, passive size-only, passive wall entries	done

Capability	Status
CVD-divergence absorption & exhaustion entries (indicator-gated)	done
Three self-contained risk scripts (basic / zone / VWAP-band)	done
trade_type column + per-finder & per-risk notes; direction flipping	done
ASSET_INFO tick injection via HIDDEN_PARAMS (30+ instruments)	done
Per-asset session times	todo
Consume big_trades_folder dataset	todo
Min-risk threshold → relocate SL / extend TP when POC too close to VAH/VAL	todo
Walk-forward validation on multi-year enriched data	todo

Performance figures

Older performance / Monte-Carlo tables described the legacy single-entry model and have been removed pending a fresh backtest. With seven finders and three risk scripts, results should be broken down by **trade_type** and **risk_script** before any numbers are quoted here.